Benchmark for MongoDB Vector Search

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This section contains the following pages, which provide information about our MongoDB Vector earch performance benchmark and how you can use it to test, evaluate, and improve your own vector search performance:

- Benchmark Overview
- Benchmark Results
- Additional Recommendations

Recall and Latency results from our MongoDB Vector Search Benchmark

How to Use This Benchmark

he primary goal for these pages is to significantly reduce friction for your first vector test at scale 10M vectors) when evaluating performance for MongoDB Vector Search.

These pages provide a set of initial configurations (embedding model dimensionality, quantization regime, `numCandidates` selection.

run tests confidently. You might need to modify your configuration based on the dataset and

query patterns relevant to your use case, as this is only meant to be a starting point.

Reading Recommendations

When reading these pages, we recommend that you focus on the primary concern that is most relevant to your use case. We provide guidance for the following primary concerns: **Recall**, **Cost**, and **Latency/Throughput**.

Use the guidance that is most appropriate for your use case:

Read these sections in the following order:

1. The following sections in the Benchmark Overview:
   - Quantization
   - Dimensionality
   - Search Node Configuration
2. The following sections in the Benchmark Results:
   - Summary of Results
   - Multidimensional Benchmark
   - Larger Benchmark
   - Recall and Latency Analysis
   - Filtering
3. Additional Performance Recommendations

Changelog

Date	Description
2025-07-21	Release of benchmark guide and results demonstrating how MongoDB Vector Search scales on a 5.5M Multidimensional and 15.3M 2048d Amazon Dataset with Voyage AI's `voyage-3-large` embeddings under various conditions

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Recall and Latency results from our MongoDB Vector Search Benchmark

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To view the full chart, see the Claude artifact.

How to Use This Benchmark

The primary goal for these pages is to significantly reduce friction for your first vector test at scale (>10M vectors) when evaluating performance for MongoDB Vector Search.

These pages provide a set of initial configurations (embedding model dimensionality, quantization regime, `numCandidates` selection, filtering criteria, Search Node configuration) that you can use to run tests confidently. You might need to modify your configuration based on the dataset and

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